

Bounds check:

When if, we use same 10 element array:

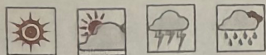
Ex) `int arr[10] = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10};`

But, we initiate $\langle \langle 2, 8 \rangle \rangle$, totally $2 \times 8 = 16$ threads. 10 threads take care of the 10 elements, what about the rest 6?

That is where a problem arises, coz there is no idle thread, it will always process it. So, at that point we will:

- * Have our other threads interfere in our workload & process.
- * Also, could cause Out of Bounds error in the future.

To avoid this we use global index of threads



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$$\text{int globalid} = \text{blockIdx.x} * \text{blockDim.x} + \text{threadIdx.x}.$$

Now, we can use this inside our Global function to prevent this case.

Our Globalid must be less than our array length, so that, other threads won't execute. (start arr)

```
-global- void mult(int *arr, int n) {
```

```
    int globalid = blockIdx.x * blockDim.x + threadIdx.x
```

```
    if (globalid < n) {
```

```
        // execute the function
    }
```

starts from 0 to 10 threads

So, it will work until thread 10 executes. After that, it won't.